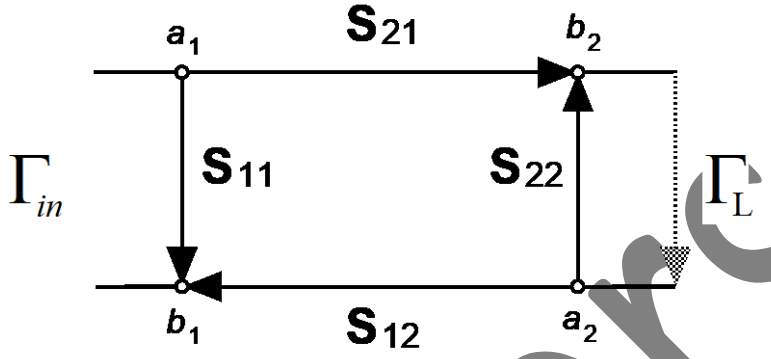
Model answer to Q.1

(a) Using a signal flow graph or otherwise, derive an expression for the input voltage-wave reflection coefficient for a 2-port network that is terminated with a load impedance Z_L in terms of the network's S-parameters.



Flow graph representation of a device with output load termination

Using Mason's non-touching loop rule it can easily be shown that:

$$\Gamma_{in} = S_{11} + \frac{S_{21}S_{12}\Gamma_L}{1 - S_{22}\Gamma_L}$$

$$where, \Gamma_L = \frac{Z_L - Zo}{Z_L + Zo}$$

[10]

(b) From the expression derived in 1(a), describe how stability circles can be created on the Smith chart. What does this stability circle represent when it encompasses the normalised impedance matched, z_0, point on the Smith chart and when it does not encompass this point?

Conditional stability occurs when the input voltage-wave reflection is equal or greater than unity. Therefore, the stability circle can be determined by equating the expression in 1(a) for the reflection coefficient to unity. When the circle encompasses the matched impedance, z_0, point then a load's voltage wave reflection coefficient is inside the circle this corresponds to unconditional stability and conditional stability outside the circle. Conversely, the opposite is true when the stability circle does not encompass the z_0 point.

[5]

(c) Given a 2-port network with the following S-parameters:

$$[S] = \begin{bmatrix} 0.5 & 0.5 \\ 10 & 0.5 \end{bmatrix} \quad (1.1)$$

What is the input voltage-wave reflection coefficient if the output port is terminated with a load impedance Z_L = -5 Ω and a reference impedance Z_0 = 50 Ω?

Answers

Γ_L = -1.222 and therefore Γ_{IN} = -3.3

[5]

(d) For the 2-port network given in 1(c) the Rollett's stability factor is 2.3. Comment on the apparent contradiction between the Rollett's stability factor and the input voltage-wave reflection coefficient found in 1(c). What is the maximum gain that can be achieved with a passive load impedance termination?

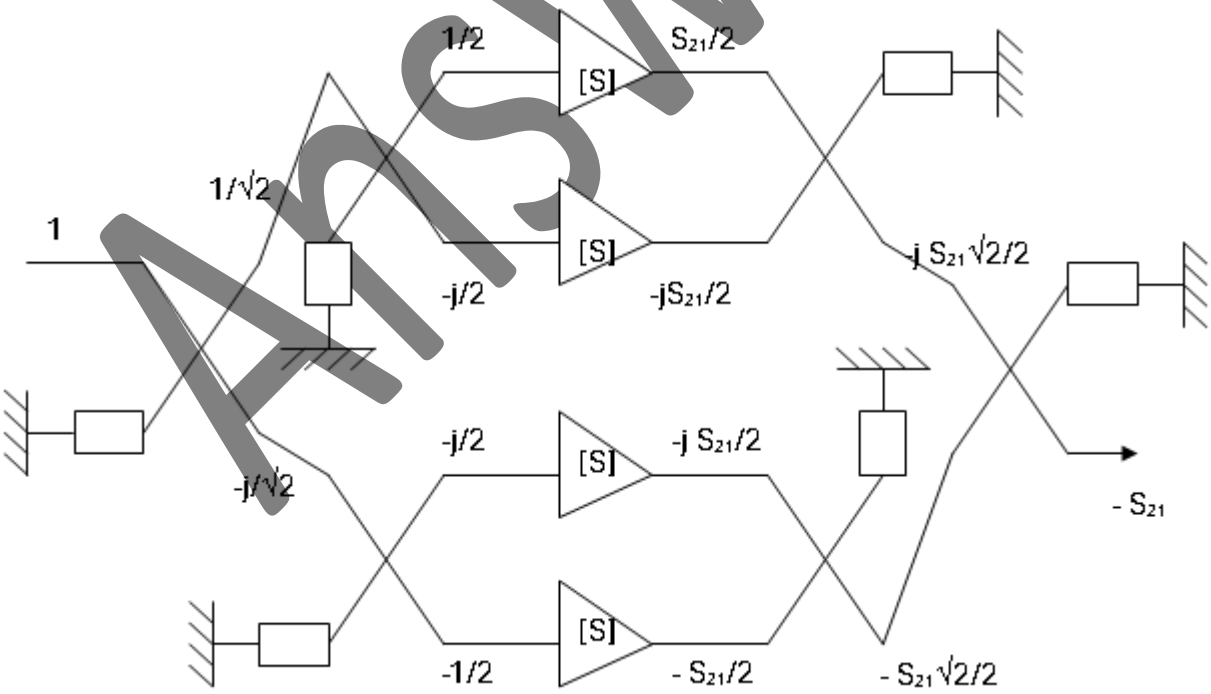
A Rollett's stability factor of K = 2.3 is greater than unity and, therefore, it would normally represent an unconditionally stable circuit. However, since the magnitude of the reflection coefficient of the load is greater than unity then the circuit is now conditionally stable.

When K is greater than unity (and the magnitudes of S_{11} and S_{22} are both less than unity), the device is unconditionally stable and the maximum gain that can be achieved is called the maximum available gain (MAG), given by:

$$\mathrm{MAG}=\left|\frac{S_{21}}{S_{12}}\right|\left(K-\sqrt{K^2-1}\right) = 4.575 = 6.6 \text{ dB}$$

[5]

(e) Draw the topology of a double-balanced amplifier.



[5]

(f) If 3 dB quadrature couplers are used in conjunction with identical non-ideal single-ended amplifiers, use S-parameter analysis to determine expressions for the overall insertion gain and input return loss. Assume the couplers are perfectly matched to the reference impedance, Z_0, and the interconnections between the main components are ideal.

Answers

$$S_{21}|_{overall} = \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right)S_{21}\left(\frac{-j}{\sqrt{2}}\right)\left(\frac{-j}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{2}}\right)\left(\frac{-j}{\sqrt{2}}\right)S_{21}\left(\frac{-1}{\sqrt{2}}\right)\left(\frac{-j}{\sqrt{2}}\right) + \left(\frac{-j}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right)S_{21}\left(\frac{-j}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right) + \left(\frac{-j}{\sqrt{2}}\right)\left(\frac{-j}{\sqrt{2}}\right)S_{21}\left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right) = S_{21}$$

$$Insertion\ Loss = 10\log\left\{|S_{21}|^2\right\}$$

$$S_{11}|_{overall} = \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right)S_{11}\left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{2}}\right)\left(\frac{-j}{\sqrt{2}}\right)S_{11}\left(\frac{-j}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right) + \left(\frac{-j}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right)S_{11}\left(\frac{1}{\sqrt{2}}\right)\left(\frac{-j}{\sqrt{2}}\right) + \left(\frac{-j}{\sqrt{2}}\right)\left(\frac{-j}{\sqrt{2}}\right)S_{11}\left(\frac{-j}{\sqrt{2}}\right)\left(\frac{-j}{\sqrt{2}}\right) = 0$$

$$Return\ Loss = 10\log\left\{|0|^2\right\} \rightarrow -\infty$$

[10]

(g) For the topology in 1(e), if the working single-ended amplifiers have a forward voltage wave transmission coefficient of $S_{21} = |10|\angle 35°$, determine the overall insertion gain and input return loss if one of the amplifiers fails, such that $S_{21} = 0$. Assume that there is no change in the input or output impedances of the failed transistor. What is the main application of this topology and what are its advantages and disadvantages when compared to a single-ended amplifier?

The main application of this amplifier is power combining, since the output power is ideally a factor of 4 greater than that of the single-ended amplifier. If one of the single-ended amplifiers fails then $S_{21}|_{overall} = 3S_{21}/4$ and Insertion loss is $10\log\{(3|S_{21}|/4)^2\} = 17.5$ dB, i.e. a drop of 2.5 dB from a fully working amplifier. The input return loss should not change if the impedance of the failed amplifier doesn't change and is, therefore, still minus infinity. Therefore, this type of power combining amplifier is useful because it provides redundancy in the case of failure and also ideal port impedance matching. The disadvantages of this topology is that it requires 4 identical single-ended amplifiers. Also, practical losses in the couplers result in a direct loss in power gain and output efficiency will be significantly reduced.

[10]

Answers

Model answer to Q.2

a) Electromagnetic radiation can cause heating of biological tissue via two main phenomena: dielectric heating and Joule heating.

i) Briefly explain the mechanism behind dielectric heating.

Dielectric heating occurs due to molecular rotation in response to an applied electromagnetic field. As molecules move to align with the applied field, they electrically interact with surrounding molecules, which dissipates part of the energy as heat, analogous to friction in a macroscopic sense.

[2]

ii) Briefly explain the mechanism behind Joule heating.

Joule heating occurs due to free electrons in a conductive material being accelerated by an applied electromagnetic field. These accelerated electrons then collide with the associated molecular lattice, converting their kinetic energy into thermal energy, dissipating the applied energy as heat.

[2]

b) The relative effective dielectric permittivity (also known as the dielectric constant) of certain biological tissues can be characterized by a Debye model. Table 2 shows the coefficients for a single-pole Debye model describing human body tissue from 1 MHz to 10 GHz:

Table 2: Single-pole Debye model coefficients.

ε_∞	σ_0 [S/m]	$\Delta\varepsilon_1$	τ_1 [s]
20	0	20	$\frac{1}{2\pi \cdot 100 \cdot 10^6}$

i) Derive an algebraic expression for the real part of the relative effective dielectric permittivity ($\varepsilon_{r\,eff}'$) for a one-pole Debye model.

From the notes:

$$\varepsilon_{r\,eff}(\omega) = \varepsilon_\infty - j\frac{\sigma_0}{\omega\varepsilon_0} + \frac{\Delta\varepsilon_1}{1 + j\omega\tau_1}$$

Rationalising the denominator for the RHS term:

$$\frac{\Delta\varepsilon_n}{1 + j\omega\tau_n} = \frac{\Delta\varepsilon_n(1 - j\omega\tau_n)}{(1 + j\omega\tau_n)(1 - j\omega\tau_n)} = \frac{\Delta\varepsilon_n}{1 + (\omega\tau_n)^2} - j\frac{\omega\tau_n\Delta\varepsilon_n}{1 + (\omega\tau_n)^2}$$

Substituting back into the expression for $\varepsilon_{r\,eff}(\omega)$:

$$\varepsilon_{r\,eff}(\omega) = \varepsilon_\infty + \frac{\Delta\varepsilon_1}{1 + (\omega\tau_1)^2} - j\left(\frac{\sigma_0}{\omega\varepsilon_0} + \frac{\omega\tau_1\Delta\varepsilon_1}{1 + (\omega\tau_1)^2}\right)$$

Therefore, the real part is:

$$\varepsilon_{r\,eff}'(\omega) = \varepsilon_\infty + \frac{\Delta\varepsilon_1}{1 + (\omega\tau_1)^2}$$

[3]

ii) Derive an algebraic expression for the imaginary part of the relative effective dielectric permittivity ($\varepsilon_{r\,eff}''$) for a one-pole Debye model.

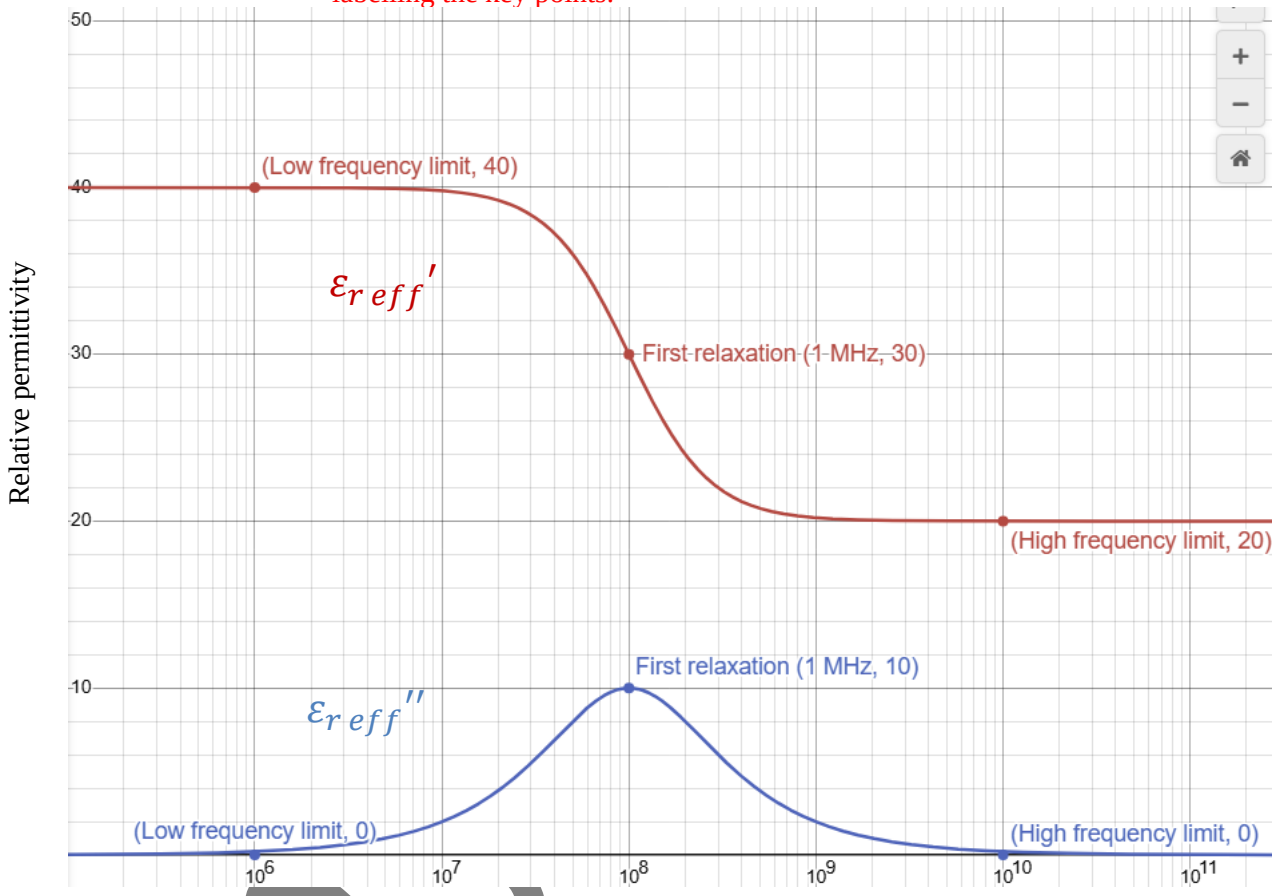
From the notes:

$$\varepsilon_{r\,eff}''(\omega) = \frac{\sigma_0}{\omega\varepsilon_0} + \frac{\omega\tau_1\Delta\varepsilon_1}{1 + (\omega\tau_1)^2}$$

Answers

[1]

iii) Sketch the real and imaginary parts of the relative effective dielectric permittivity using the data from Table 2 on a log-frequency axis, labelling the key points.



Answers

- Correct shape of relaxations for real part.
- Correct shape of relaxations for imaginary part.
- Low limit for real part.
- Low limit for imaginary part.
- High frequency limit for real part.
- High frequency limit for imaginary part.
- Relaxation point for real part.
- Relaxation point for imaginary part.

[8]

c) A mobile phone is undergoing SAR testing for use in the United States. The peak average SAR is measured to be 3 W/kg, averaged over a volume of 1.92 cm^3.

i) What is the FCC limit on average SAR?

From the notes: The FCC limit on average SAR is 1.6 W/kg averaged over 1g of tissue.

[2]

ii) The tissue has a mass volume density of 1,040 kg/m^3. What is the mass that the average SAR has been averaged over?

1.92 cm^3 = 1.92 x 10^{-6} m^3. Therefore, the averaged mass = 1.92 x 10^{-6} x 1,040 x 1,000 = 1.9968 g, or approx. 2 g.

[2]

iii) Give two reasons why this mobile phone may not comply with the FCC limit. *Hint: Consider the averaging volumes or masses.*

This mobile phone does not unconditionally comply with the FCC limit because there is not enough information about the SAR distribution to convert between the averaging volumes/masses.

A 2 g average has less resolution than a 1 g average and as a result may miss (or blur-out) any SAR hotspots that a 1 g average may not.

[2]

iv) State a scenario in which absorbed electromagnetic power can be SAR-compliant but still be harmful to humans.

High-power pulsed RF.

[1]

v) Calculate the maximum theoretical temperature change, assuming constant irradiation for 10 mins, given a specific heat capacity of 3,600 J/kg/°C.

From the notes: $\Delta T\ [°C] = \mathrm{SAR} \cdot \frac{t}{c_p}$ Therefore, ΔT = 3 x 60 x 10 / 3,600 = 0.5°C

[2]

Answers

Model answer to Q.3

(a) Given a square wave signal generator, with a clock frequency that can vary between zero and 1.0 GHz, propose how this could be used to generate a sinusoidal signal at a frequency of 1.8 GHz. *Hint: Consider the Fourier series of the square wave.*

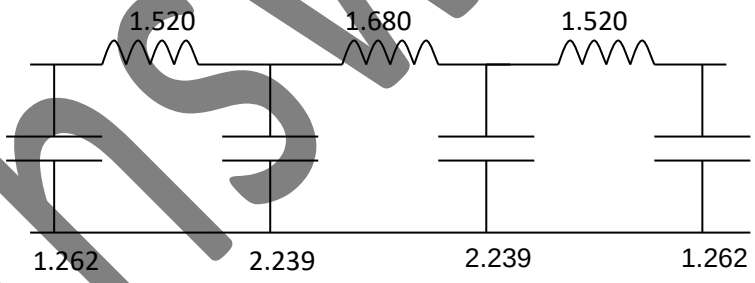
Given a square wave signal generator, with a clock frequency that can vary between zero and 1000 MHz, set the output to 600 MHz. Using a band-pass filter having a centre-frequency set to the third harmonic of the signal generator, the desired 1800 MHz sinusoidal signal can be extracted. Alternatively, a high-pass filter can be used, however, the 5th and 7th etc., harmonics will also be present but at much lower power levels.

[5]

(b) Design a lumped-element *L*-*C* high-pass filter to meet the following specifications:

Pass band attenuation ripple	0.1 dB
-3 dB cut-off frequency:	1,500 MHz
Stop band frequency:	600 MHz
Stop band attenuation:	> 70 dB
Source impedance, Z_S:	50 Ω
Load impedance, Z_L:	50 Ω

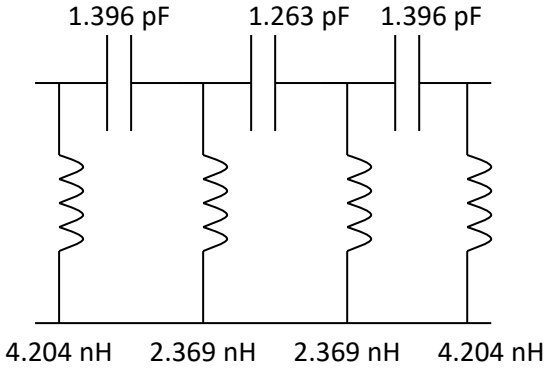
From graphs provided, for Chebyshev filters with a 0.1 dB ripple, a 7th order filter is required to achieve an out-of-band rejection of > 70 dB with an fc/f ratio of 1500/600 = 2.5.
From tables provided, with $R_S = R_L = Zo = 50$, the prototype low-pass filter and associated coefficients are given below:

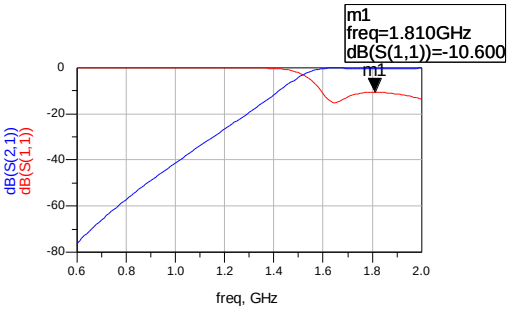


High-pass de-normalising:

Shunt inductor: $L_P = \dfrac{R_L}{2\pi f_c Ln}$

Series Capacitor: $C_S = \dfrac{1}{2\pi f_c CnR_L}$





Answers

The slight deviation in the results from those predicted by theory are due to limited component tolerances used within the simulations.

[10]

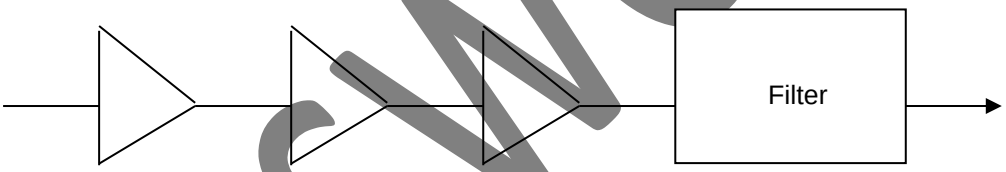
(b) Determine the worst-case levels of return losses within both the pass band and the stop band for the filter in 3(b). How could the stop band return loss adversely affect the implementation of 3(a)? Suggest a suitable topology for overcoming this problem.

The worst-case level of return loss within the pass band is 10 log[1- antilog(-0.1/10)] = -16.43 dB. This will have no effect within the pass band. However, well-below the pass band the return loss will be 0 dB. At the fundamental frequency of 600 MHz, all the signal will be reflected back by the filter and the square wave signal generator may not work properly (if at all). One possible solution is to insert a circulator between the square-wave generator and the filter, however, passive circulators tend to be narrow band in nature. However, a more practical solution is to use a balance topology for the filters, whereby two identical filters are embedded between two identical ultra-broadband 3 dB quadrature directional couples. This approach is inherently impedance matched across the bandwidth of the couplers.

[10]

Model answer to Q.4

An amplifier chain is illustrated in Figure 4.1. All sub-systems are assumed to be perfectly impedance matched.



G = 15 dB G = 8 dB G = 4 dB G = -0.7 dB

IP3 = 30 dBm IP3 = 40 dBm IP3 = 50 dBm

F = 1 dB F = 2 dB F = 3 dB

Figure 4.1

For an overall input power of +3 dBm, calculate the following at the output of each sub-system:-

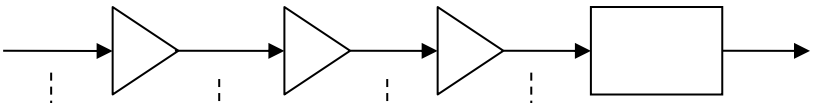
Using the following general equations:

$$C = P_{OUT}(fo) = G_1 G_2 P_{IN} \quad and \quad IP_3 = \left(IP_3|_1 \; G_2\right) \| \, IP_3|_2$$

$$IMD_3 = \frac{C}{I_3} \sim \left(\frac{IP_3}{C}\right)^2 \equiv 2\left(IP_3[dBm] - C[dBm]\right)[dBc]$$

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \frac{F_4 - 1}{G_1 G_2 G_3} + \ldots$$

Answers



G [dB]		+15	+8	+4	-0.7	
IP_3 [dBm]		+30	+40	+50		
F [dB]		+1	+2	+3		
C= Pout [dBm]	+3	+18	+26	+30	+29.3	[2]
IP_3 [dBm]		+30	+35.9	+39.5	+38.8	[8]
IMD_3 [dBc]		+24	+19.8	+19	+19	[3]
I_3 [dBm]		-6	+6.2	+11	+10.3	[2]
F [dB]					+1.1	[10]

All variables have their usual meaning.

Answers